

Lab 01: Installation & Lab Environment Setup

Cybersecurity F26

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1. Introduction

This exercise introduces the setup of the lab environment that will be used throughout the course. Students will install and configure Kali Linux as the primary penetration testing platform. Kali includes a wide range of security assessment tools that will be used in subsequent lectures and lab exercises.

In addition, the Metasploitable 2 and Metasploitable 3 virtual machines will be deployed as intentionally vulnerable target systems. Although these systems are outdated, the fundamental penetration testing techniques required to identify, exploit, and analyze their vulnerabilities remain highly relevant.

Upon completing this setup, students will have a fully functional practice environment and will be able to further develop their skills by exploring additional publicly available penetration testing labs and challenges.

2. Preparing the Lab Environment

Notes and Requirements

- **Disk Space:** At least 100 GB of free disk space is required for this lab. This includes the Kali Linux virtual machine (with databases), Metasploitable 2 (Linux), and Metasploitable 3 (Windows) images.
- **Additional Tools:** Next lab exercise will also make use of OpenVAS (Greenbone Security Assistant) and Nessus for vulnerability assessment, and snort for intrusion detection. These tools will be installed and configured in the subsequent lab.
- **Security Warning:** The Metasploitable virtual machines contain intentionally vulnerable and untrusted software. Treat these systems as hostile and ensure they remain strictly isolated from your host system and external networks. Virtual machines are designed for this purpose. Use them accordingly.

Virtualization Platform

- This lab assumes the use of VirtualBox as the virtualization platform. This choice is based on its availability across major operating systems, not on technical superiority.
- Students are free to use alternative solutions such as VMware, which may offer improved performance or features depending on the host system.

Network Configuration

- For the network setup (see section 4 below), you may choose a more restrictive configuration or a simpler setup that prioritizes functionality.
- From a pen tester's perspective, ensuring reliable communication between virtual machines is the primary practical concern.

Virtual Machine Setup

- All Metasploitable virtual machine images are available via the Cybersecurity F26 resource folder provided in [NextCloud](#).
- At a minimum, the lab environment should include three virtual machines:
 - Two target machines running Metasploitable 2 & 3
 - One attacker machine, running Kali Linux, from which all testing and exploitation activities will be conducted.

Notes on the Windows Subsystem for Linux (WSL)

- **WSL as an Installation Option:** WSL can be used to install Kali Linux. In particular, WSL 2 provides a full virtualization environment by running a real Linux kernel. However, by default, WSL is tightly integrated with the host Windows system. While this integration offers convenience, it also reduces isolation between the attacking environment and the primary work system. For this reason, we may prefer a dedicated virtualization solution, which provides stronger separation and better reflects real-world penetration testing setups.
- **Graphical User Interfaces (GUI):** Graphical user interfaces are supported when using WSL. Native GUI support is included starting with Windows 11. On earlier versions of Windows, an external X11 server must be installed and configured manually to enable graphical applications.

Notes on Different CPU Platforms

- **Platform Compatibility:** Students must ensure platform compatibility when setting up the lab environment. All virtual machine images provided via [NextCloud](#), and the Metasploitable images in general, are built for the amd64/x86 architecture.

- **Intel/AMD Systems:** This architecture is supported natively on most Windows and Linux systems and should not present issues for the majority of users.
- **Apple Silicon (M1/M2/M3):** Modern Apple systems using Apple Silicon (ARM architecture) are not natively compatible with x86-based Metasploitable images. While Kali Linux is available in ARM-compatible versions and runs efficiently on these systems, equivalent Metasploitable images are not readily available. Students using Apple Silicon systems must choose one of the following alternatives:
 - Use hardware emulation to run x86 virtual machines (with reduced performance),
 - Work in collaboration with a teammate who has an Intel/AMD-based system,
 - Use a cloud-based solution.

3. Installing Virtual Machines for the Lab

Installing VirtualBox

Linux Users

- Install VirtualBox using your distribution package manager (e.g., apt, dnf, pacman). This ensures proper integration with your operating system and simplifies updates.

Windows and macOS Users

- VirtualBox must be installed manually by downloading the appropriate installer.
 - **Download VirtualBox:** Visit the [VirtualBox](https://www.virtualbox.org/) website download the installer that matches your operating system under “VirtualBox Platform Packages.”
 - **Install VirtualBox:** Run the downloaded installer and follow the installation instructions.
 - **Install the VirtualBox Extension Pack:** On the same webpage, locate the section titled “VirtualBox Extension Pack.”
 - i. Click “All supported platforms” to download the .vbox-extpack file.
 - ii. Double-click the downloaded file to install the extension pack. VirtualBox will automatically launch and complete the installation.
 - iii. The Extension Pack is required for full functionality, including USB support and improved device handling.

Installing Kali Linux

- Download the Virtual Machine Image

- **Preferred option:** Download the updated Kali Linux VM image from the course [NextCloud](#) share.
- **Alternative option:** Download the official Kali Linux VM image from the Kali website.

When doing so, ensure that you select the VirtualBox image and the Kali Linux version is compatible with your operating system and CPU architecture.

- Documentation (Official Image Only)
 - If you are using the official Kali image, follow the installation instructions provided in the [official Kali documentation](#).
- Import the Virtual Machine
 - For preconfigured (cloud/OVA) images: In VirtualBox, select File → Import Appliance, or double-click the downloaded image file
 - This will open the import dialog, allowing you to review and import the virtual machine configuration.
- Configure Memory (RAM)
 - Select the Kali Linux VM in the VirtualBox main window.
 - Click Settings → System.
 - Adjust the Base Memory (RAM):
 - 1 GB is the minimum and may be sufficient for basic tasks.
 - Allocating more RAM is recommended for better performance, provided your host system has enough memory.
- Configure CPU
 - In Settings → System → Processor:
 - The default configuration assigns 2 CPUs.
 - 1 CPU is sufficient for most exercises, but additional CPUs may improve performance.
 - Do not allocate more CPUs than your host system can comfortably support.
- Network Settings
 - To enable communication between VMs, refer to section 4 for detailed options.
 - For a simple and functional setup:
 - Go to Settings → Network
 - Set “Attached to” to NAT Network
 - This configuration allows internet access and inter-VM connectivity.
- Finalize Setup

- Click OK to save your settings.
- Double-click the Kali Linux VM to start it.
- Default Credentials
 - The default credentials for Kali VM images are:
Username: kali
Password: kali
 - I will recommend you change the default password.
- Keeping Kali Up to Date
 - Ensure Kali Linux is fully updated by running the following commands in a terminal:
`sudo apt update`
`sudo apt full-upgrade -y`
- Using a Different Linux Distribution (Optional)
 - If you choose to use a Linux distribution other than Kali, you must manually install the required security tools. At a minimum, ensure the following are available: nmap, netcat, metasploit, whois, host, etherape, pnsnscan, legion, nessus, mysql, john (John the Ripper), openvas / Greenbone Security Assistant. **Note:** This tool list may be updated during the course.

Installing Metasploitable 2

Metasploitable 2 & Metasploitable 3 are intentionally vulnerable and contain untrusted software by design. Treat these virtual machines as hostile and ensure they are strictly isolated from your host system and any production networks.

- Download the Metasploitable 2 from either [Rapid7](#) or the one provided in [NextCloud](#)
- Unzip in the desired directory and import the .vmdk file from VirtualBox.
- Importing the Virtual Machine
 - In VirtualBox, select File → Import Appliance, or double-click the .vmdk file.
 - Follow the import wizard to complete the installation.
- Default Login Credentials
 - The default credentials for the Metasploitable 2 image are:
Username: msfadmin
Password: msfadmin
- Network Configuration
 - Configure the network settings according to section 4 or use NAT Network for a simple setup.

Installing Metasploitable 3

- Downloading the Metasploitable 3
 - Download the image from either [VagrantUp](#), or the [NextCloud](#) via Vagrant provided earlier.
- Preparing the File
 - After downloading, unzip the archive.
 - Rename the extracted file so that it ends with .ova. Technically, the file is a tar archive; changing the file extension allows Windows and VirtualBox to recognize it correctly. VirtualBox filters importable files by extension, and .ova files are listed automatically in the import dialog.

Alternatively: You may also clear the file type filter in the VirtualBox import dialog and manually select the unzipped file. Renaming the file to .ova.gz will make it appear in the list, but VirtualBox will not allow the import to proceed.

- Follow the same import procedure as Metasploitable 2 and install the .ova file.
- Default login credentials for the Metasploitable 3 are:

Username: **vagrant**

Password: **vagrant**

- Finally, configure the network settings the same as Metasploitable 2.

NOTE: If installing Metasploitable3 from command prompt, you will have to install vagrant plugin via the command: `vagrant plugin install vagrant-reload`

- Then install vb-guest for vagrant using: `vagrant plugin install vagrant-vbguest`
- And finally, Metasploitable 3 in vagrant using:
`vagrant box add rapid7/metasploitable3-win2k8`

IMPORTANT: If you get any error message while installing, it is an issue with one of the plugins. To rectify, navigate to the path below from the user folder:

`.vagrant.d/gems/3.3.8/gems/vagrant-vbguest-0.32.0/lib/vagrant-vbguest/hosts/virtualbox.rb`

- Edit the last file 'virtualbox.rb' in notepad.
- Line 84 'path && File.exists' should be 'path && File.exist' without an s.
- You might also see another error with 'MountDiskImage' which has no any effect on the machine.

4. Setting Up Inter-VM Communications

To perform penetration testing between the two virtual machines, Kali Linux (attacker) and Metasploitable (target)—both systems must be able to communicate with each other. This requires placing them on the same virtual network.

This section provides a simple network configuration in VirtualBox, suitable for most lab exercises, and a more sophisticated setup, which provides stronger isolation and is preferable when working with untrusted or third-party virtual machines.

- VirtualBox Networking Options

Table 1 summarizes the available VirtualBox networking modes and the types of connectivity they allow between virtual machines and external networks.

	VM ↔ Host	VM1 ↔ VM2	VM → Internet	VM ← Internet
HostOnly	Yes	Yes	No	No
Internal	No	Yes	No	No
Bridged	Yes	Yes	Yes	Yes
NAT	No	No	Yes	Port forwarding
NAT Network	No	Yes	Yes	Port forwarding

Table 1. VirtualBox networking options and supported interconnections (adapted from [1]). More on VirtualBox Networking (see the manual from [2])

- Recommended Network Configuration

For this set-up, two configurations are relevant:

- NAT Network (Simple Setup): The NAT Network (NATNet) option is the simplest configuration. It allows internet access for virtual machines, automatic IP address assignment via DHCP, and communication between all VMs connected to the same NAT Network

The downside is limited control over which machines can communicate with each other. This option is sufficient for most introductory lab exercises and is recommended if you want to get the environment working quickly.

- Internal Network + NAT (Advanced Setup): A more controlled and secure approach that combines: Internal Networking for inter-VM communication, and A separate NAT adapter for controlled internet access.

This setup allows you to restrict communication strictly to selected virtual machines, better isolate vulnerable targets from external networks, and model

¹ [VirtualBox Forums: How to communicate two VMs](#)

² [VirtualBox Networking](#)

more realistic network topologies. This configuration is preferable when working with untrusted VMs or when stronger isolation is required.

- **Notes on Other Virtualization Solutions**

- VMware: Similar functionality can be achieved using VMware's virtual networking features. However, DHCP configuration may require additional setup and is not covered here.
- WSL 2: When using WSL 2, Host-Only Networking is required to allow communication with Metasploitable virtual machines. However, due to the tight integration between WSL and the host system, running intentionally vulnerable machines such as Metasploitable in WSL is strongly discouraged. A dedicated virtualization environment provides significantly better isolation and security.

NAT Network: The Easy Option

The simplest way to enable communication between virtual machines is to use NAT Network (NATNet). This option allows all connected virtual machines to access the internet through the host system and communicate with each other automatically via DHCP.

- **Creating a NAT Network in VirtualBox**
 - In VirtualBox, select File → Tools → Network Manager
(or File → Preferences → Network, depending on your VirtualBox version).
 - Click Add to create a new network named NatNetwork.
 - Verify that DHCP is enabled for this network.

This ensures that IP addresses are assigned automatically to the virtual machines.

- **Assigning the NAT Network to the Virtual Machines**
 - Select the Kali Linux VM in the VirtualBox main window.
 - Click Settings → Network.
 - Set Attached to: NAT Network.
 - Select Name: NatNetwork (or the name you created earlier).
 - Click OK to save the settings.
 - Repeat the same steps for the Metasploitable 2 & 3 VMs, ensuring that it is connected to the same NAT Network.
 - **Note:** Virtual machines must be powered off to modify network settings.

Internal Network: The More Complex Option

A more sophisticated networking setup uses the Internal Network option. Virtual machines connected to an internal network do not have direct internet access, which is often desirable, especially for intentionally vulnerable systems like the Metasploitable 2 & 3.

- IP Address Configuration Options
 - To enable communication over an Internal Network, one of the following must be configured:
 - i. Assign static IP addresses to both virtual machines, or
 - ii. Run a DHCP server on the Kali VM, or
 - iii. Enable a DHCP server in VirtualBox (recommended)
- Configuring a VirtualBox DHCP Server
 - These steps must be performed from the command line on the host system. Windows users may find it convenient to change to the directory containing the VBoxManage binary to avoid specifying full paths.
 - i. List existing internal networks:
`VBoxManage list intnets`
 - ii. VBoxManage existing DHCP servers:
`VBoxManage list dhcpservers`
 - iii. Create and enable DHCP server for the internal network
`VBoxManage dhcpserver add`
`--network=CyberSecLab \`
`--lowerip 172.30.1.20 \`
`--upperip 171.30.1.50 \`
`--netmask 255.255.255.0 \`
`--ip 172.30.1.1 \`
`--enable`

This configuration creates a DHCP server for the internal network named CyberSecLab, assigning IP addresses from the 172.30.1.0/24 range. NOTE: The backslash (\) indicates that the command continues on the next line. Also, remember to choose CyberSecLab when configuring the internal network adapter.

- Verifying the Configuration
 - Start all three virtual machines
 - Log in to the Metasploitable 2 and Metasploitable 3 VMs using their default credentials.
 - Verify that the machines have received an IP address and are reachable from Kali